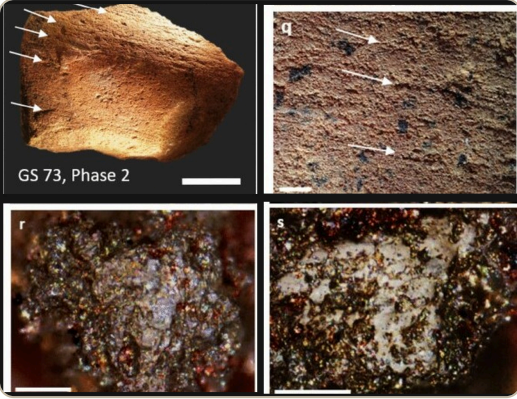


How one stone becomes evidence

Read across for twelve months of growth. **Read down** from the marked phrase, to what it proves, to the continuum descriptor. Only the phrases that prove a criterion are marked — the black text is the glue.



What does GS73 suggest about life at Madjedbebe in ancient Australia?

ANCIENT OBJECT GS73, a sandstone millstone fragment excavated from square B5/52 at Madjedbebe on Mirarr Country.	MODERN RECORD Hayes and colleagues published photographs, micrographs and laboratory results in a peer-reviewed study in 2022.	EXCAVATION CONTEXT Phase 2, modelled to 68.7–50.4 thousand years ago. The earliest end of this range is debated.
MEASURED EVIDENCE Deep partial grooves; levelled quartz grains; striations; bright polish on the raised parts of grains.	RESIDUES 41 starch grains recovered and plant-related compounds detected. The exact plant was not identified.	KNOWN LIMITS Two chemical peaks were identified as plastic contamination. The integrity of the Phase 2 layer is contested.

■ Origin, features & purpose <u>marked like this</u>	▲ Historical context <u>marked like this</u>	● Historical interpretations <u>marked like this</u>	◆ Accuracy, usefulness & reliability <u>marked like this</u>
--	--	--	--

YEAR LEVEL	Grade 5	Year 6	Year 7 EXPECTED	Year 8	Year 9	Year 10
WORKED EXAMPLE	GS73 is a stone source found at Madjedbebe. I can see deep grooves and shiny patches . It comes from ancient Australia and may have been used for grinding. Researchers disagree about exactly how old its soil layer is . The stone is useful for showing tool use , but it cannot tell us the user’s name .	GS73 is a stone tool : a broken piece of a grinding stone from Madjedbebe, on Mirarr Country. Photographs show deep grooves and shiny, polished patches . People were living there many thousands of years ago , and they used stones like this one to grind their food . Some archaeologists believe the layer is about 65,000 years old . Others believe the soil has moved, so the date is not certain . The stone is useful evidence that people were grinding materials , but it cannot tell us which plants they were grinding .	The ancient object is GS73, a broken piece of a sandstone grinding stone excavated at Madjedbebe, on Mirarr Country . The modern source is a set of photographs published by Hayes and her team in 2022 . Their purpose was to record the wear and argue that people ground food there for a very long time . In ancient Australia, grinding allowed people to eat hard seeds and tough plants . It also shows they understood the resources around them . Hayes and her team interpret this as plant processing , because the stone carries deep grooves, microscopic polish and 41 starch grains . Other researchers question how old the layer really is . GS73 is very useful for studying how people prepared food , because its shape, its wear and its plant traces all agree , but it cannot tell us which plant was ground .	GS73 is a broken millstone recovered from Phase 2, one of the oldest occupation layers at Madjedbebe . Hayes and her team photographed it and examined it under a microscope . They published the images so that other researchers could inspect the same evidence and judge their interpretation . Madjedbebe is a rock shelter on Mirarr Country that was occupied for many thousands of years . The grinding marks show that people were already preparing plant foods early in Aboriginal history . Hayes explains the wear and the plant residues as signs of seed grinding . Clarkson argues that the artefacts remained where they were dropped, which supports the early date . Williams points to termite activity and moving sediment, which would make that date less reliable . The shape, the microscopic wear, the starch and the plant chemicals all point in the same direction , so the grinding interpretation is well supported. Even so, I would ask whether the residues could be contaminated, and whether the layer could have shifted .	GS73 is material evidence, excavated from a recorded layer called Phase 2 . Hayes and her team published photographs, microscope images and laboratory results together . They did this because one form of evidence alone would not convince a scientific audience . The study belongs to wider research into the deep history of Aboriginal Australia , and it was carried out with the permission of Mirarr custodians . Hayes analyses the grooves, the polish and the plant residues as evidence of seed processing . Clarkson analyses the condition of the site and the dating to defend the early chronology . Williams uses signs of termite disturbance and sediment movement to challenge it . Because the wear and the residues agree independently, grinding is the interpretation I would trust most . The exact plant and the precise age are weaker, because both depend on preservation and on how stable the deposit is .	GS73 is an ancient artefact, but we meet it only through a modern scientific publication. Its explicit evidence is the recorded wear and the plant residues preserved on the stone . Its implicit argument is that grinding technology in Australia is far older than earlier accounts allowed , and that argument is aimed at other archaeologists . The investigation reflects modern laboratory testing carried out with Mirarr custodians . It also reveals what those researchers value: testing claims, seeking permission, and stating honestly how certain they are . Hayes and Clarkson accept the early chronology because several independent records converge on it . Williams gives more weight to termite disturbance and sediment movement . Their interpretations differ because they evaluate the formation of the site differently . Overall, GS73 is strong evidence for grinding, but only qualified evidence for a precise age . Its main strength is that separate tests agree ; its limits are the unidentified plant, the modern plastic contamination, and the ongoing argument about the layer .
WHAT IT PROVES	■ Names the source and two visible features: ‘stone source’, ‘deep grooves’ and ‘shiny patches’. Origin and purpose are not yet developed. ▲ Places GS73 broadly in ‘ancient Australia’, but gives little detail about life or practice. ● Notices disagreement about the date without describing the competing arguments. ◆ Gives one use and one limit: tool use is visible, but the user cannot be named.	■ ‘Archaeological artefact’, ‘sandstone grinding-stone fragment’ and ‘grooves and polish’ identify type and features; purpose is not yet explained. ▲ ‘A very early period of Aboriginal occupation’ identifies the context and connects it to repeated tool use. ● Recognises two positions on the date, but does not yet describe the evidence behind them. ◆ Recognises usefulness for grinding and a clear limit: the material processed cannot yet be identified.	■ Separates the ancient object from the modern 2022 record, then identifies origin, features and why the figure was created. ▲ Outlines how grinding supported plant preparation and knowledge of local resources. ● Describes the plant-processing interpretation and the separate challenge to the earliest date. ◆ Concludes that converging methods make plant processing useful evidence, then bounds the claim with two limits.	■ Describes the object and explains how the modern figure lets readers inspect its form, wear and residues. ▲ Describes Madjedbebe, Mirarr Country and grinding within deep Aboriginal history. ● Explains the debate by connecting each interpretation to a different evidence base. ◆ Compares independent evidence for function, then asks targeted reliability questions about contamination and layer movement.	■ Explains why the peer-reviewed figure combines several records, and which audience and argument it addresses. ▲ Explains the deep-history research context and the role of Mirarr custodial permission. ● Analyses each interpretation through its evidence: wear and residue, site integrity and dating, or termite and sediment movement. ◆ Identifies grinding as a stronger claim than exact plant or exact age, and explains why.	■ Analyses explicit evidence, the publication’s implicit argument, its purpose and its scientific audience. ▲ Uses context to explain the creators’ values: empirical testing, custodial permission and careful qualification. ● Exposes why interpretations differ — researchers give different weight to converging records and site-formation evidence. ◆ Weighs a strength, three limitations, and two claims that carry different degrees of certainty.
LEARNING CONTINUUM	GRADE 5	YEAR 6	YEAR 7	YEAR 8	YEAR 9	YEAR 10
ORIGIN, FEATURES & PURPOSE	I can name a historical source and a feature I notice.	I can identify different types of historical sources.	I can identify the origin, content features and purpose of historical sources.	I can describe the origin, content features and purpose of historical sources.	I can explain the origin, content features and purpose of historical sources.	I can analyse the origin, explicit and implicit meaning, and purpose of historical sources.
HISTORICAL CONTEXT	I can say where or when a source comes from.	I can identify the historical context of a source.	I can outline the historical context of a source.	I can describe the historical context of sources.	I can explain historical context, including important events that influenced a source’s creation.	I can explain historical context to identify the motivations, values and attitudes of a source’s creator.
HISTORICAL INTERPRETATIONS	I can notice that people may explain the past differently.	I can recognise that historians may interpret historical events and developments differently.	I can describe different historical interpretations of the past.	I can explain different historical interpretations and contested debates about the past.	I can analyse historical interpretations and contested debates and identify the evidence used to support them.	I can evaluate historical interpretations and contested debates, including why or how interpretations differ.
ACCURACY, USEFULNESS & RELIABILITY	I can say one thing a source can and cannot tell us.	I can recognise that some historical sources are more useful than others.	I can draw conclusions about the usefulness of sources.	I can compare and contrast historical sources and ask questions about their accuracy, usefulness and reliability.	I can compare and contrast historical sources and identify their accuracy, usefulness and reliability.	I can compare and contrast sources and analyse their accuracy, usefulness and reliability.

Source: panels adapted from Figure 3 of Hayes, E. H. et al. (2022), “65,000-years of continuous grinding stone use at Madjedbebe, Northern Australia,” *Scientific Reports* 12, 11747, doi:10.1038/s41598-022-15174-x (open access, CC BY 4.0). Figure prepared by Elspeth H. Hayes. The dating comparison uses Clarkson et al. (2017) and Williams et al. (2021). **The research was conducted with permission from the custodians of Madjedbebe, May Nango and Djaykuk Djandjomerr, and Mirarr Senior Traditional Owner Yvonne Margarula, with Gundjeihmi Aboriginal Corporation facilitating the work.** Mirarr-related images and information were published for general educational purposes; no commercial use is authorised without prior Mirarr consent. See SOURCE-NOTICE.md.